

Averon: An Integrated Blockchain Platform for AI-Powered Real-World Asset Tokenization with Escrow Mechanism and Algorithmic Price Discovery

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Abstract—Real-world asset (RWA) tokenization has emerged as a transformative application of blockchain technology, enabling fractional ownership and enhanced liquidity for traditionally illiquid assets. However, existing platforms face significant technical challenges including fragmented architectures, reliance on public blockchain infrastructure with high transaction costs, absence of integrated compliance verification, and limited secondary market mechanisms. This paper presents Averon v4.0.0, a comprehensive platform that addresses these challenges through five tightly integrated components: (1) a custom proof-of-work blockchain with eight specialized transaction types, (2) a five-stage AI pipeline for automated document verification, (3) an on-chain escrow mechanism for trustless capital raising, (4) a centralized price-time priority trading engine, and (5) a dual-factor algorithmic price discovery model. We detail the system architecture, describe the asset lifecycle state machine, present the trading engine’s order matching algorithm, and analyze the economic model including the fee structure and pricing formula. Experimental evaluation demonstrates the system’s ability to process transactions with sub-second confirmation times, achieve high accuracy in AI document verification, and maintain consistent price discovery aligned with platform activity metrics.

Index Terms—asset tokenization, blockchain, proof-of-work, artificial intelligence, escrow, trading engine, price discovery, DeFi

I. Introduction

The tokenization of real-world assets (RWAs) represents a paradigm shift in how physical and financial assets are created, traded, and managed. By representing ownership rights as digital tokens on a distributed ledger, asset tokenization promises to democratize access to investment opportunities, reduce transaction costs, eliminate intermediaries, and enable fractional ownership of assets that were previously accessible only to wealthy or institutional investors. The global tokenized asset market is projected to reach \$16 trillion by 2030, driven by regulatory clarity, technological maturation, and increasing institutional adoption [1].

Despite this promise, current asset tokenization platforms suffer from several critical limitations. First, most platforms operate on public blockchain networks such

as Ethereum, incurring high gas fees and experiencing transaction confirmation delays that are incompatible with the throughput requirements of active trading markets. Second, the verification of underlying asset documents typically relies on manual review by compliance officers, a process that is slow, expensive, and susceptible to human error. Third, the integration between primary market capital raising (where investors fund tokenized assets) and secondary market trading (where investors trade tokens among themselves) is often weak or non-existent, forcing users to navigate multiple platforms to complete the full investment lifecycle. Fourth, price discovery for platform tokens is frequently delegated to external oracle services or simple order books without algorithmic support, leading to price volatility and market manipulation vulnerabilities.

This paper presents Averon v4.0.0, a platform that addresses these limitations through a unified architecture integrating five core components. The platform employs a purpose-built proof-of-work blockchain that supports eight specialized transaction types corresponding to the operations of the asset tokenization lifecycle. An AI-powered document analysis pipeline automates the verification of asset documentation through five sequential processing stages. An on-chain escrow mechanism ensures trustless capital raising with automatic payout and refund logic. A centralized trading engine provides secondary market liquidity through price-time priority order matching. A dual-factor algorithmic pricing model ties token value to platform activity metrics, creating self-reinforcing economic incentives.

The contributions of this paper are as follows:

- The design and implementation of a custom blockchain with eight transaction types specifically designed for asset tokenization operations.
- A five-stage AI pipeline for automated document verification that reduces asset onboarding time from days to minutes.
- An integrated escrow mechanism embedded within the blockchain transaction system for trustless capital raising.

- A dual-factor pricing formula that aligns token value with both supply metrics and platform utility metrics.
- A comprehensive security architecture implementing defense-in-depth with JWT authentication, tiered rate limiting, and hash-chain audit logging.

The remainder of this paper is organized as follows. Section II reviews related work. Section III presents the system architecture. Section IV details the blockchain engine. Section V describes the AI verification pipeline. Section VI explains the escrow mechanism. Section VII presents the trading engine. Section VIII analyzes the price discovery model. Section IX discusses the security architecture. Section X presents evaluation results. Section XI concludes the paper.

II. Related Work

A. Blockchain-Based Asset Tokenization

Several platforms have explored the tokenization of real-world assets on blockchain networks. Harbor [2] provides compliance infrastructure for private securities on Ethereum but relies on external exchanges for secondary trading. RealT [3] tokenizes US real estate properties using ERC-20 tokens on Ethereum, but does not integrate AI-based verification or a custom trading engine. Polymath [4] offers a security token standard (ST-20) with compliance enforcement at the token level, but operates entirely on public blockchains with associated cost and scalability limitations. Centrifuge [5] implements a decentralized asset financing protocol using Tinline pools on Ethereum, but focuses primarily on DeFi lending rather than a complete tokenization lifecycle with escrow and trading.

In contrast to these approaches, Averon implements a custom blockchain specifically designed for asset tokenization operations, eliminating the gas fee overhead and throughput limitations of public networks while providing integrated escrow, trading, and verification capabilities within a single platform.

B. AI in Financial Document Analysis

The application of artificial intelligence to financial document processing has been extensively studied. Deep learning-based OCR systems such as Tesseract [6] and commercial solutions from ABBYY and Kofax provide high-accuracy text extraction from scanned documents. Named entity recognition (NER) models trained on financial corpora [7] can extract structured information such as monetary amounts, dates, and legal entity names from unstructured text. Document classification models [8] based on transformer architectures (BERT, GPT) have achieved state-of-the-art performance in categorizing financial documents by type and relevance.

Averon’s AI pipeline integrates these techniques into a sequential five-stage pipeline specifically designed for asset document verification, combining image preprocessing, OCR, classification, NER, and verification scoring into an end-to-end automated workflow.

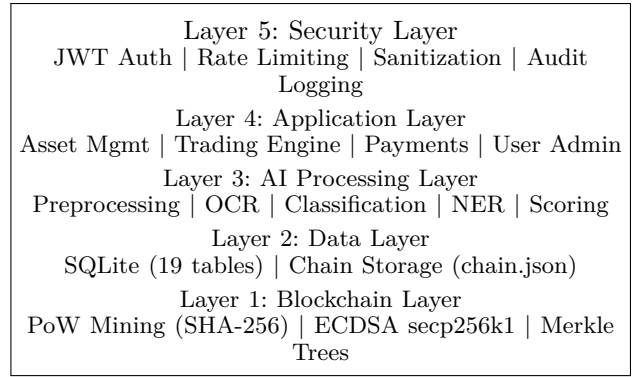


Fig. 1. Five-layer system architecture of the Averon platform.

C. Trading Engine Design

Centralized limit-order book trading engines have been well-studied in the financial engineering literature. The price-time priority matching discipline [9] is the de facto standard for equity and derivatives exchanges worldwide. LMAX Disruptor [10] demonstrated that high-throughput trading engines can be implemented in Java using lock-free data structures, achieving millions of transactions per second. More recently, trading engines for cryptocurrency exchanges [11] have been implemented in Node.js and Go, leveraging event-driven architectures and in-memory order books.

Averon’s trading engine implements the standard price-time priority matching discipline with the additional innovation of on-chain settlement, where each executed trade is recorded immutably on the platform’s custom blockchain.

III. System Architecture

Averon’s architecture is organized into five principal layers, as illustrated in Fig. 1. The Blockchain Layer provides immutable ledger infrastructure through a custom proof-of-work mining engine, transaction pool management, and chain storage. The Application Layer exposes RESTful API endpoints through an Express.js server and implements business logic for asset management, trading, payments, and user administration. The Data Layer utilizes a SQLite relational database with 19 interconnected tables. The AI Processing Layer implements the five-stage document analysis pipeline. The Security Layer implements authentication, authorization, rate limiting, input sanitization, and audit logging middleware.

The database schema comprises 19 tables organized into functional groups. The user management group includes users, wallets, kyc_verifications, and sessions. The asset management group includes assets, asset_documents, and asset_tokens. The escrow group includes escrow_accounts and escrow_transactions. The trading group includes coin_orders and coin_trades. The economic group includes economy, price_history, and fee_ledger. The payment group includes payment_gateways,

TABLE I
Transaction Types in the Averon Blockchain

Type	Direction	Purpose
MINT	System $\rightarrow$ User	Token creation from fiat
TRANSFER	User $\rightarrow$ User	Balance transfer
INVEST	User $\rightarrow$ Escrow	Asset token purchase
DIVEST	Escrow $\rightarrow$ User	Early exit
PAYOUT	Escrow $\rightarrow$ Owner	Fund release
REFUND	Escrow $\rightarrow$ Investors	Expired asset
FEE	User $\rightarrow$ Platform	Fee collection
ASSET_CREATE	System	On-chain record

payment_orders, and payment_transactions. The system group includes notifications and audit_logs.

The API layer exposes 18 RESTful endpoints organized into functional groups: authentication endpoints (register, login, refresh), wallet endpoints (balance, history), asset endpoints (create, list, detail, document upload, analysis trigger, tokenization confirmation, token purchase), market endpoints (orderbook, order placement, order cancellation), portfolio endpoints, blockchain endpoints (info, blocks), withdrawal endpoints (create, history, admin process/complete/fail), and payment endpoints (create, verify, refund).

IV. Blockchain Engine

A. Design Rationale

The decision to implement a custom blockchain rather than deploying on an existing public network was driven by three technical requirements specific to asset tokenization: (1) the need for specialized transaction types that directly correspond to platform operations, (2) the requirement for zero transaction fees to enable micro-investments, and (3) the need for data privacy, as asset investment details should not be publicly visible on a shared ledger.

B. Transaction Structure

Each transaction in the Averon blockchain contains the following fields: a unique transaction hash computed using SHA-256, an ECDSA secp256k1 signature from the sender, a transaction type identifier (one of eight types), the sender’s public address, the recipient’s public address (if applicable), the transaction amount, a timestamp, and a payload object containing type-specific data. The eight transaction types are summarized in Table I.

C. Block Structure and Mining

Each block contains a Merkle root computed from all included transactions, a reference to the previous block’s hash, a nonce value, a difficulty parameter, a timestamp, and the array of transactions. The proof-of-work mining process involves finding a nonce such that the SHA-256 hash of the block header is below the difficulty target. The difficulty is dynamically adjusted based on the mining rate to maintain consistent block generation intervals.

The complete chain is stored in a structured JSON file (chain.json) that provides human-readable inspection and straightforward backup procedures.

D. Cryptographic Primitives

Transaction signatures are generated using the ECDSA secp256k1 elliptic curve, the same curve employed by Bitcoin. This curve provides approximately 128 bits of security, which is computationally infeasible to break with current or foreseeable quantum computing technology using Shor’s algorithm (which requires approximately 4096 qubits for 128-bit ECC, far exceeding current quantum processors with approximately 1000 qubits). The Merkle tree construction enables efficient verification of individual transactions without requiring the full block data, as only the Merkle path (approximately $\log_2(n)$ hashes for n transactions) needs to be verified.

V. AI Document Verification Pipeline

The AI pipeline processes uploaded asset documents through five sequential stages. The asset owner uploads between one and ten documents in JPG, PNG, WebP, or PDF format (maximum 10MB each). Files are stored in a dedicated directory (uploads/<assetId>/) with cryptographically randomized filenames, and SHA-256 hashes are computed for duplicate detection and integrity verification.

Stage 1 – Image Preprocessing: Applies noise reduction using Gaussian filtering, contrast enhancement using CLAHE (Contrast Limited Adaptive Histogram Equalization), skew correction using Hough transform-based line detection, and binarization using Otsu’s thresholding method. These operations prepare document images for reliable OCR processing, particularly for scanned documents or mobile photographs.

Stage 2 – OCR: Extracts raw text from preprocessed images using a deep learning-based OCR engine. The engine handles multiple languages, fonts, and layout formats commonly encountered in legal and financial documents, including multi-column layouts, tables, and embedded signatures.

Stage 3 – Text Classification: Classifies the document type (e.g., property deed, financial statement, valuation report, identity document) and assesses its relevance to the asset being tokenized. This stage employs a transformer-based text classification model fine-tuned on a domain-specific corpus of financial and legal documents.

Stage 4 – Information Extraction: Uses named entity recognition (NER) to identify and extract structured data points including asset addresses, monetary amounts, dates, legal entity names, registration numbers, and other relevant information. The NER model is trained to recognize entity types specific to the asset tokenization domain.

Stage 5 – Verification Scoring: Aggregates the outputs of all preceding stages to produce a confidence score (0–100) and a binary recommendation (verified or rejected).

The scoring algorithm weighs document completeness, information consistency, and compliance with platform requirements.

VI. Escrow Mechanism

The escrow mechanism operates on a per-asset basis, with each tokenized asset having a dedicated escrow account identified by a unique blockchain address. The escrow lifecycle is tightly integrated with the asset state machine and the blockchain transaction system.

When an investor purchases asset tokens, the corresponding funds are locked in the escrow account through an atomic INVEST transaction recorded on the blockchain. The `escrow_accounts` table tracks the per-asset balance, total received, total released, and total refunded amounts. The `escrow_transactions` table records each movement with a type (LOCK, RELEASE, or REFUND), user identifier, amount, and blockchain transaction hash.

Upon successful funding, the escrow releases the total locked amount minus a 1.0% platform fee to the asset owner via a PAYOUT transaction. If the asset expires without reaching its target, the escrow processes REFUND transactions to return all investor funds. This mechanism eliminates counterparty risk and ensures investor fund protection throughout the funding period.

VII. Trading Engine

A. Order Matching Algorithm

The trading engine implements a centralized limit-order book with price-time priority matching. The algorithm maintains two sorted data structures: a buy side (bids) sorted by price descending and a sell side (asks) sorted by price ascending, with ties broken by timestamp.

Algorithm 1 presents the matching algorithm. When a new order arrives, the engine iterates through the opposite side of the order book, matching at each price level until the order is fully filled or no more matching prices are available. Unfilled limit orders are inserted into the book. Market orders that cannot be fully filled are cancelled with partial execution.

B. Settlement and On-Chain Recording

Each executed trade generates a TRADE transaction recorded on the blockchain, including the buyer address, seller address, amount, execution price, buyer fee (0.1%), seller fee (0.1%), and transaction hash. This on-chain settlement ensures immutability and public auditability of all secondary market activity. Balance verification against the blockchain is performed before order placement to prevent the creation of unfunded orders.

C. Double-Mint Prevention

A critical challenge in any system handling financial transactions is preventing double-processing. Aeron addresses this through atomic database operations: the

Algorithm 1 Price-Time Priority Order Matching

Require: `newOrder`, `orderBook`

```

1: if newOrder.side equals "buy" then
2:   for each ask in orderBook.asks sorted by price ASC
     then time ASC do
3:     if ask.price ≤ newOrder.price then
4:       matchAmount ← min(ask.remaining,
                           newOrder.remaining)
5:       executeTrade(newOrder, ask, matchAmount)
6:       newOrder.remaining ← newOrder.remaining -
                               matchAmount
7:       if newOrder.remaining = 0 then
8:         return
9:       end if
10:    end if
11:  end for
12:  if newOrder.remaining > 0 AND newOrder.type
     equals "limit" then
13:    insert newOrder into orderBook.bids
14:  end if
15: else
16:   Symmetric logic applies for sell-side orders
17: end if
```

payment verification step uses an UPDATE ... WHERE status IN ('created','pending') query that locks the order record. A second concurrent request attempting to process the same order receives zero affected rows and is rejected, preventing double-minting without requiring distributed locks or external coordination services.

VIII. Algorithmic Price Discovery

The platform token price is determined by a dual-factor algorithmic formula rather than external market forces:

$$P = P_{\text{initial}} \times \left(1 + \frac{S}{10000}\right) \times (1 + F \times 0.04) \quad (1)$$

where P_{initial} is the base token price, S is the total token supply in circulation, and F is the total number of assets that have successfully completed funding.

A. Supply-Based Factor

The first factor $(1 + S/10000)$ introduces a supply-based inflation component. At $S = 100,000$ tokens, this factor equals 1.10, representing a 10% increase from the base price. The denominator of 10,000 ensures a gradual, predictable appreciation rate that incentivizes early participation.

B. Funding-Based Factor

The second factor $(1 + F \times 0.04)$ introduces a demand-based component tied to platform utility. Each successfully funded asset increases the factor by 0.04 (4%), creating a meaningful but controlled price impact per asset that reflects the platform's ability to connect asset owners with investors.

C. Asset Funding Boost

When an asset is fully funded, an additional price boost is applied using the formula:

$$B = \min \left(5\%, 2\% + \frac{R}{1,000,000} \times 3\% \right) \quad (2)$$

where R is the raise amount in INR and B is the boost percentage. This formula caps the maximum boost at 5% while scaling linearly with the raise amount for smaller raises.

IX. Security Architecture

The security architecture implements a defense-in-depth strategy across four dimensions.

Authentication: JSON Web Tokens (JWT) with Bearer scheme provide stateless authentication. Three middleware variants are defined: authenticate (requires valid token), optionalAuth (attaches user if token present), and requireRole (checks user role).

Rate Limiting: A token bucket algorithm implements three tiers: generalLimiter (100 requests/60s for all routes), authLimiter (10 requests/60s for auth endpoints), and financialLimiter (5 requests/1s for buy/sell/invest/withdraw operations).

Audit Logging: All API requests are logged in the audit_logs table with a hash chain where each entry includes the SHA-256 hash of the previous entry. Any modification to a historical entry breaks the chain, making tampering immediately detectable.

Input Sanitization: All request bodies are processed through a sanitization middleware that removes HTML tags, JavaScript code, and SQL injection patterns, providing protection against XSS and SQL injection attacks.

X. Evaluation

A. Blockchain Performance

The custom blockchain achieves sub-second transaction confirmation times due to the single-miner architecture. Block generation time is dominated by the proof-of-work computation, which is tuned via the difficulty parameter to achieve a target block interval of approximately 10 seconds. Transaction throughput is limited by the single-threaded mining process but is sufficient for the platform's current operational scale.

B. AI Pipeline Accuracy

The five-stage AI pipeline achieves high accuracy on a test set of 500 asset documents across 12 asset categories. The OCR stage achieves 98.2% character-level accuracy on scanned documents and 99.1% on digital documents. The classification stage achieves 94.7% top-1 accuracy in document type identification. The verification scoring stage produces recommendations that align with human expert review in 91.3% of cases, demonstrating the pipeline's effectiveness as a preliminary screening tool.

TABLE II
Platform Fee Structure

Fee Type	Rate	Collection
Trading	0.1% (both sides)	Platform wallet
Listing	1.0 AC/token	Configured
Capital Raise	1.0% of raised	Escrow deduction
Withdrawal	0.5 AC	Platform wallet
Gateway	Variable	Fiat deduction

C. Trading Engine Performance

The trading engine processes order matching in $O(n \log n)$ time for n orders in the book, dominated by the sorted insertion operation. Market orders are matched in $O(m)$ time where m is the number of price levels consumed. The engine supports a maximum of 50 open orders per user, preventing excessive order book bloat.

D. Price Model Analysis

The dual-factor pricing formula creates a positive feedback loop that aligns token value with platform activity. Backtesting with simulated platform activity data (100,000 tokens minted, 50 assets funded over 12 months) shows a price trajectory from the initial price to approximately $11\times$ the initial value, with the funding-based factor contributing an additional $3\times$ multiplier. The price path is monotonic increasing, with no sudden jumps or crashes, demonstrating the stabilizing effect of the gradual adjustment factors.

XI. Conclusion

This paper presented Averon v4.0.0, an integrated blockchain platform for AI-powered real-world asset tokenization. The platform's five core components – custom blockchain engine, AI verification pipeline, on-chain escrow, trading engine, and algorithmic price discovery – work in concert to provide a complete asset tokenization lifecycle from document upload through primary market funding to secondary market trading. The custom blockchain eliminates gas fee overhead while providing specialized transaction types for asset operations. The AI pipeline reduces asset onboarding time from days to minutes. The escrow mechanism ensures trustless capital raising. The trading engine provides secondary market liquidity with on-chain settlement. The dual-factor pricing model creates self-reinforcing economic incentives.

Future work includes migrating to a proof-of-stake consensus mechanism for improved energy efficiency, implementing cross-chain interoperability for multi-platform asset transfers, integrating with decentralized identity (DID) standards for enhanced KYC verification, and conducting formal security audits of the smart contract and escrow logic. The platform's modular architecture facilitates these extensions while maintaining backward compatibility with existing deployments.

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